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ABAP Enhancement Framework

Deloitte Technology Academy (DTA)

Agenda

Topic	Descriptions	Duration
Enhancement Framework	Introduction, Components and Technology, Enhancement Options, Enhancement Spots, Enhancement Implementations, Explicit Enhancement Options, and Implicit Enhancement Options	4 hours

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Learning Objectives

By the end of this session, you will be able to:

- Describe the new enhancement framework
- Implement implicit enhancement
- Implement explicit enhancement

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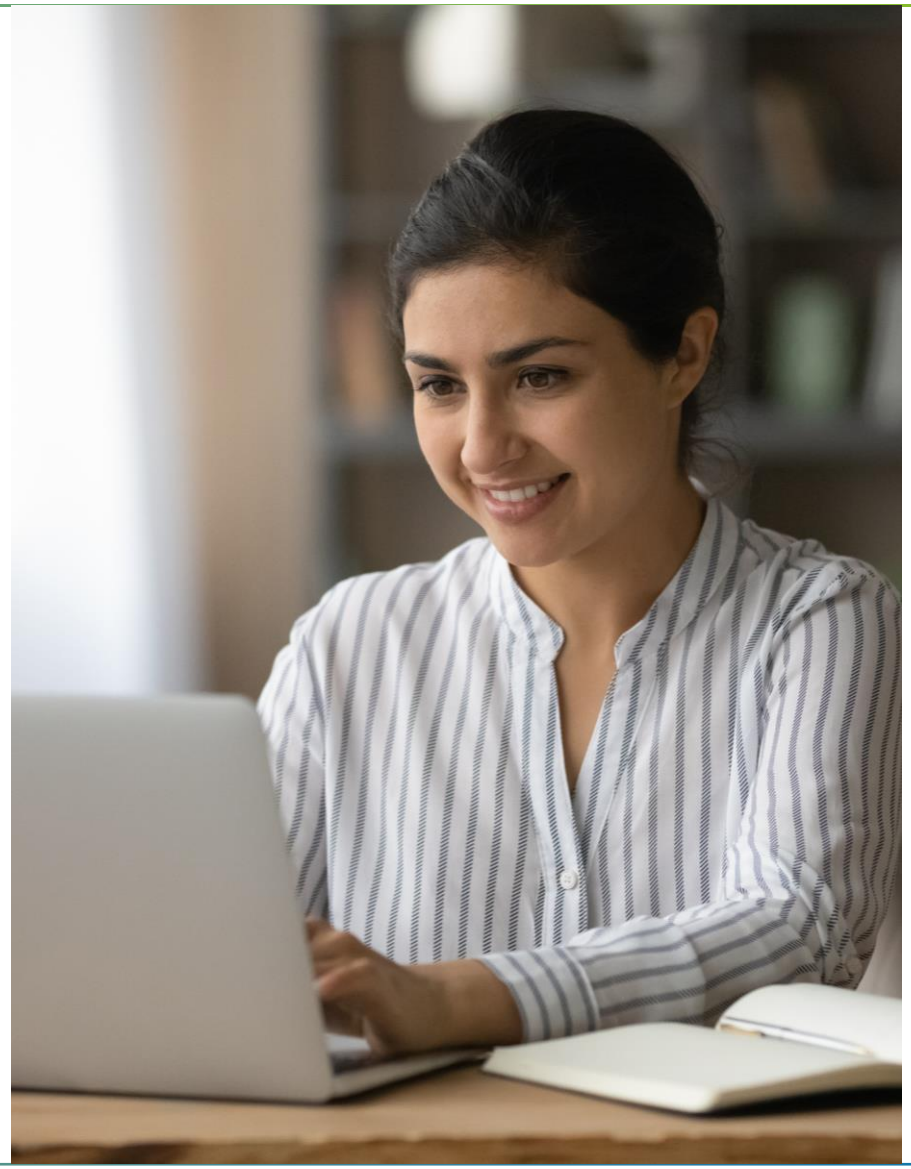
Enhancement Framework

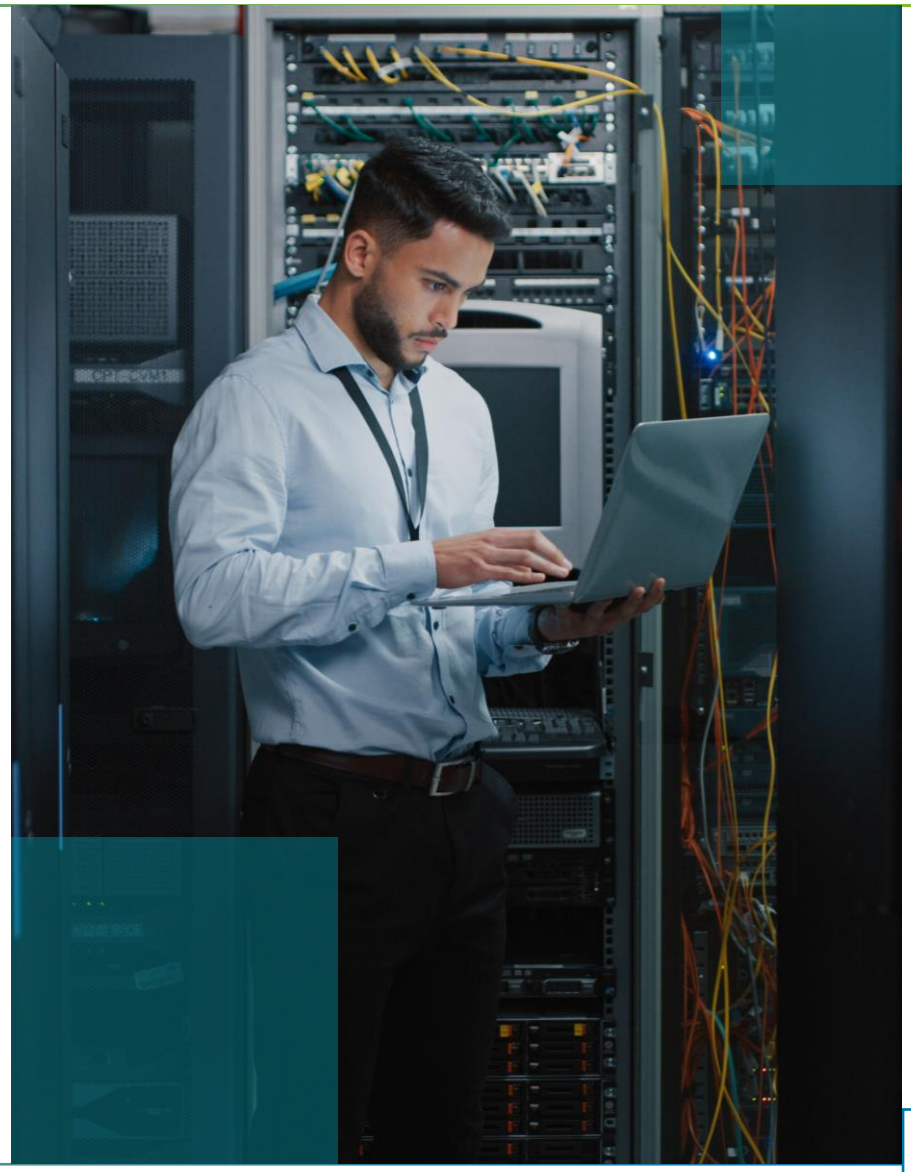


Enhancement Framework—Technologies

The enhancement framework is the new paradigm to bring all enhancement techniques under one roof.

The following are different enhancement technologies available under this framework.

- Source code enhancement
- Function group enhancement
- Class enhancement



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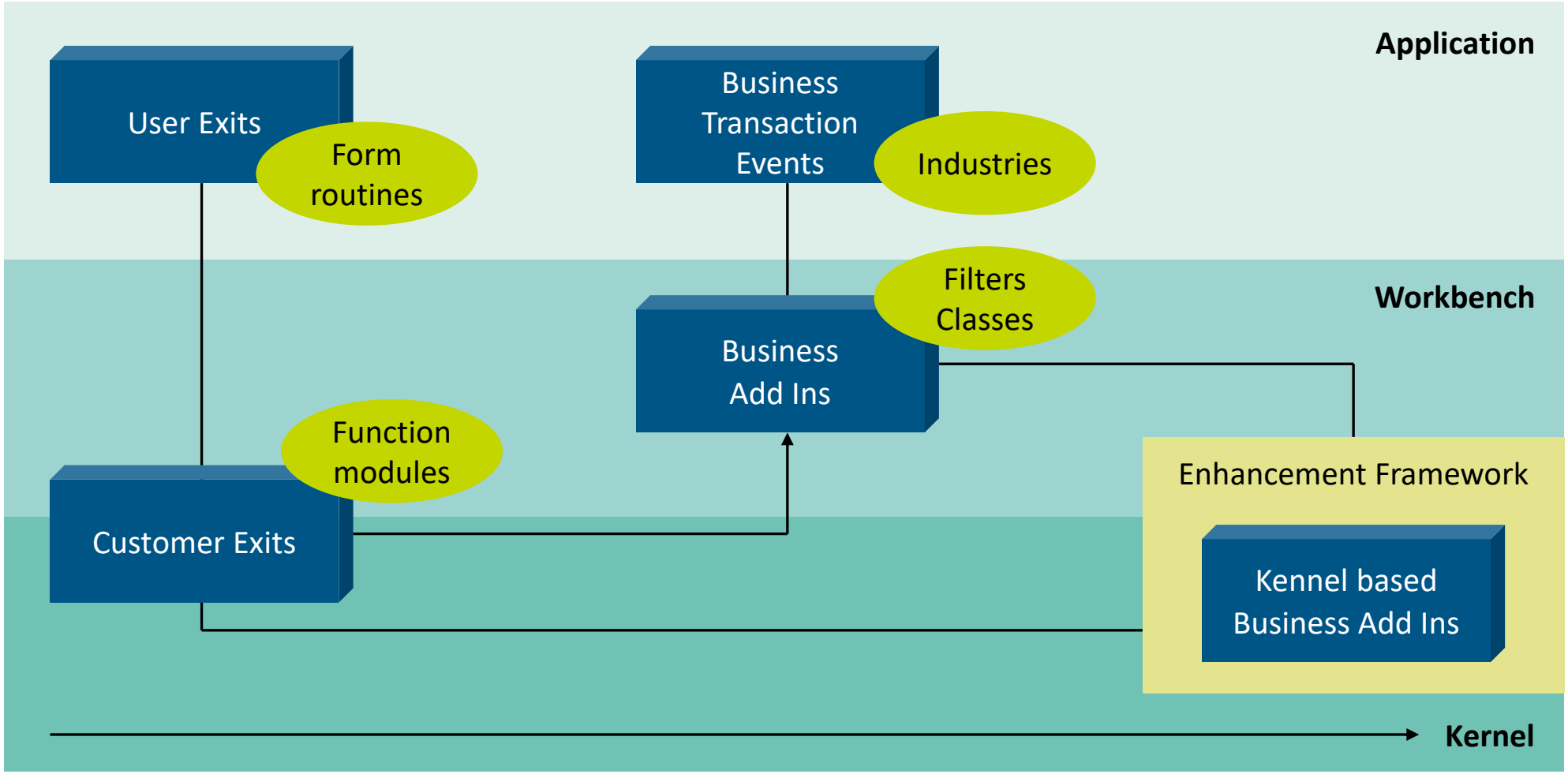
Evolutions of SAP Enhancement Technology



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Older Enhancement Techniques

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User-Exits

- The very first mechanism provided by SAP to execute custom code in between the standard SAP control flow.
- This is implemented as a subroutine call (Perform xxx).
- A classical example of user-exit is MV45AFZZ included in the order processing module of SAP R/3.

Customer Exits

- It is implemented using function modules and so has a well-defined parameter interface.
- Since the custom coding is done as part of these customer exits and is located away from the original SAP code, the maintenance is easier than user exits. (Before 4.6, only through customer exits)

BADIs

- It exists in pre-NW04s releases and is now called old classic-BADIs.
- This was the first object-oriented way to enhance the ABAP system. This, to a certain extent, allows multiple implementations with limited filter support.
- The classic-BADIs are implemented using ABAP objects.

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Drawbacks of Previous Enhancement Techniques

No support for multiple users or projects

No support for parallel developments

Will appear much more often in adjustment tools

Higher adjustment effort (during the upgrade and SP import)

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Components of Enhancement Framework

The enhancement framework consists of three higher-level components:

Enhancement Option

- Are positions in repository objects where enhancements can be made
- These options are either explicitly defined or exist implicitly

Enhancement Spots

- Enhancement spots are container objects and are used to manage explicit enhancement options
- Enhancement spots carry information about the positions at which enhancement options were created

Enhancement Implementation

- Enhancement Implementations are container objects that manage the enhancement options, both explicit and implicit

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What Are Enhancement Spots?

- Enhancement spots are created to manage explicit enhancement options.
- An explicit enhancement option is always referenced using the spot name and option name.
- An enhancement spot can contain multiple enhancement options
- An enhancement option can be assigned to multiple enhancement spots.
- A composite enhancement spot can hold multiple simple enhancement spots.
- Spots are always provided by the application developer
- No spots are required for implicit enhancement options

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Types of Enhancement Options

Enhancement options can be classified into two categories:

Explicit enhancement option	Implicit enhancement option
• Are defined explicitly by the developer at valid source code locations • Managed using enhancement spots • Enhanced using enhancement implementations • Are of two types: – Source code plug-in – Kernel BADI (or new BADI)	• Always exist and do not have to be specified by the developer • No enhancement spots are necessary • Enhanced using enhancement implementations • Are of three types: – Specific options in programs – Parameter interfaces of function modules – Methods of classes

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What Is an Enhancement Implementation?

- Used to manage enhancements
- Enhancement Implementations are created by the customer
- An enhancement option can be implemented in multiple enhancement implementation
- An enhancement implementation can contain multiple enhancements
- An individual enhancement inside an implementation is known as an enhancement implementation element
- An enhancement implementation element for an explicit enhancement option must be associated with an enhancement spot
- A composite enhancement implementation can contain multiple simple enhancement implementations

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Source Code Plug-In

The following enhancement options are available in the source code:

Enhancement—Point	Enhancement—Section
• Can be static (data declaration) or dynamic (coding) • Used as a placeholder to place additional coding • Can belong to multiple enhancement spots • Multiple active implementations are possible and will be executed with no guarantee in the order of execution	• Can be static or dynamic • Used to modify/replace a certain section of code. Only one is active at a time • Can belong to multiple enhancement spots • Implementation gets executed and the original code does not get executed

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Source Code Plug-In—Editor Modes

The following enhancement options are available in the source code:

Creating enhancement options

- Use the change mode for creating enhancement points or sections
  - Use the button to switch between change ↔ display modes

Implementing enhancement options

- Use the enhancement mode to create implementations
  - Use the button to get into the enhancement mode
  - Use the button to leave the enhancement mode

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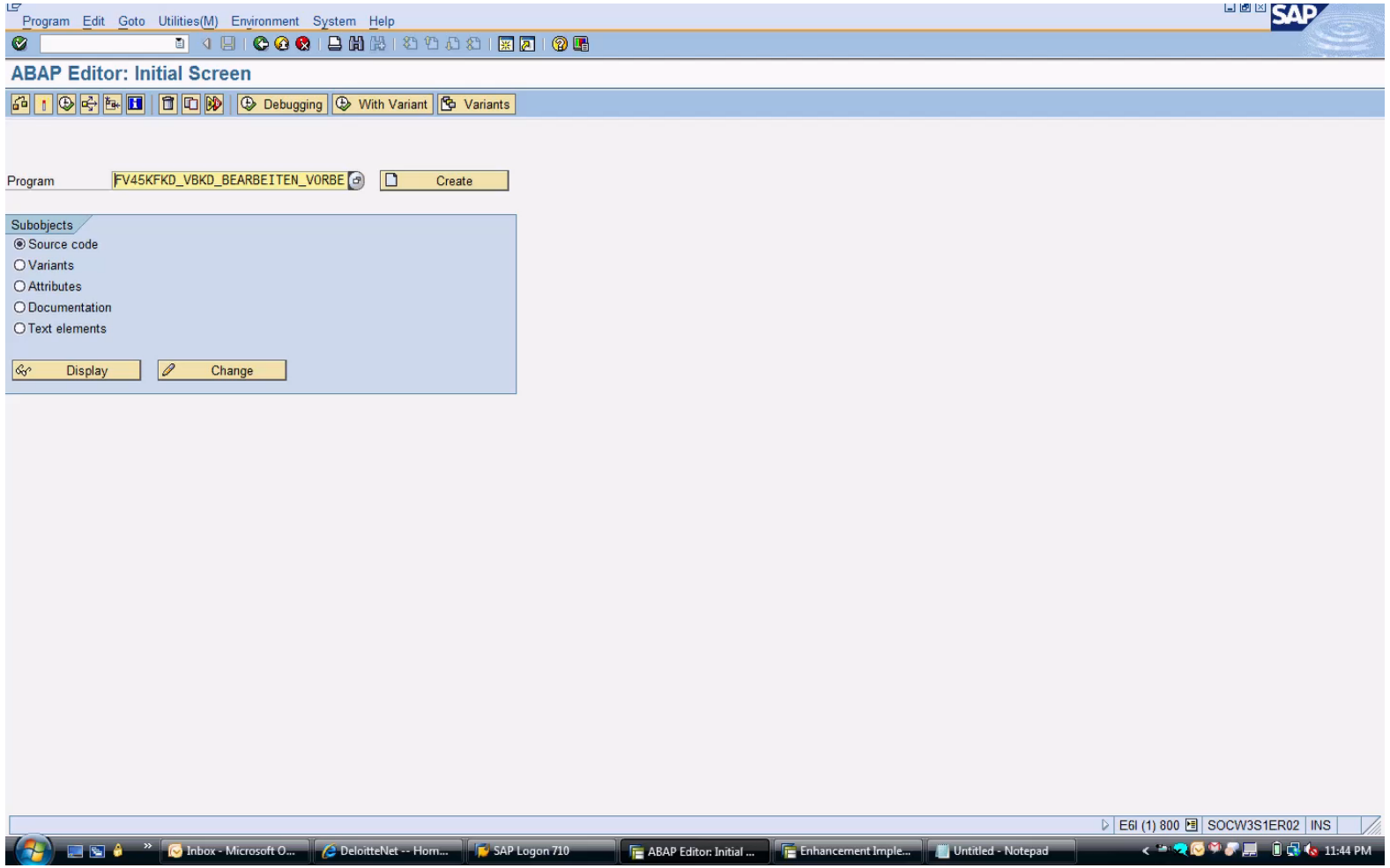
Explicit Enhancement Options—Enhancement Point: Demo



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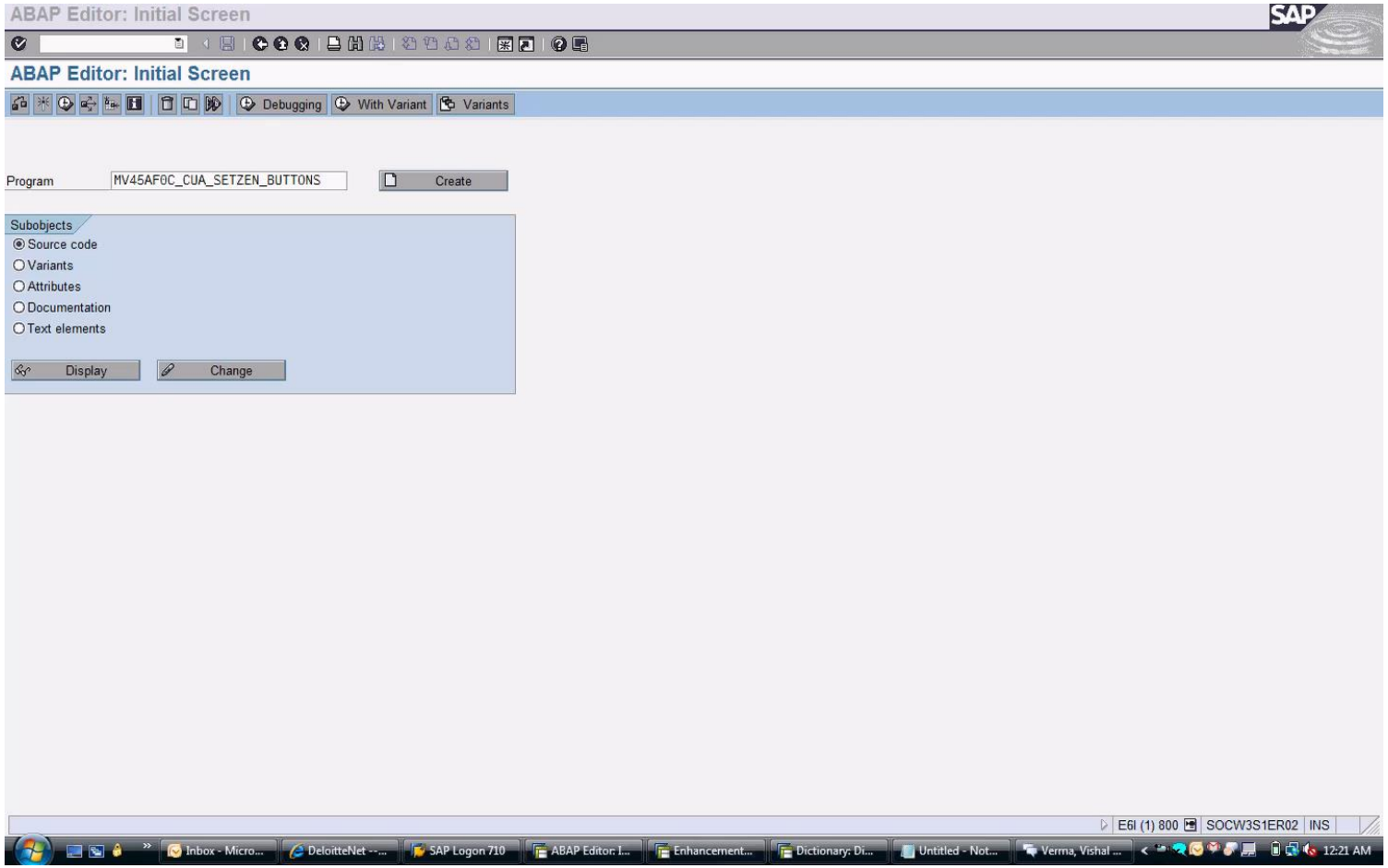
Enhancement Section—Before Implementation: Demo



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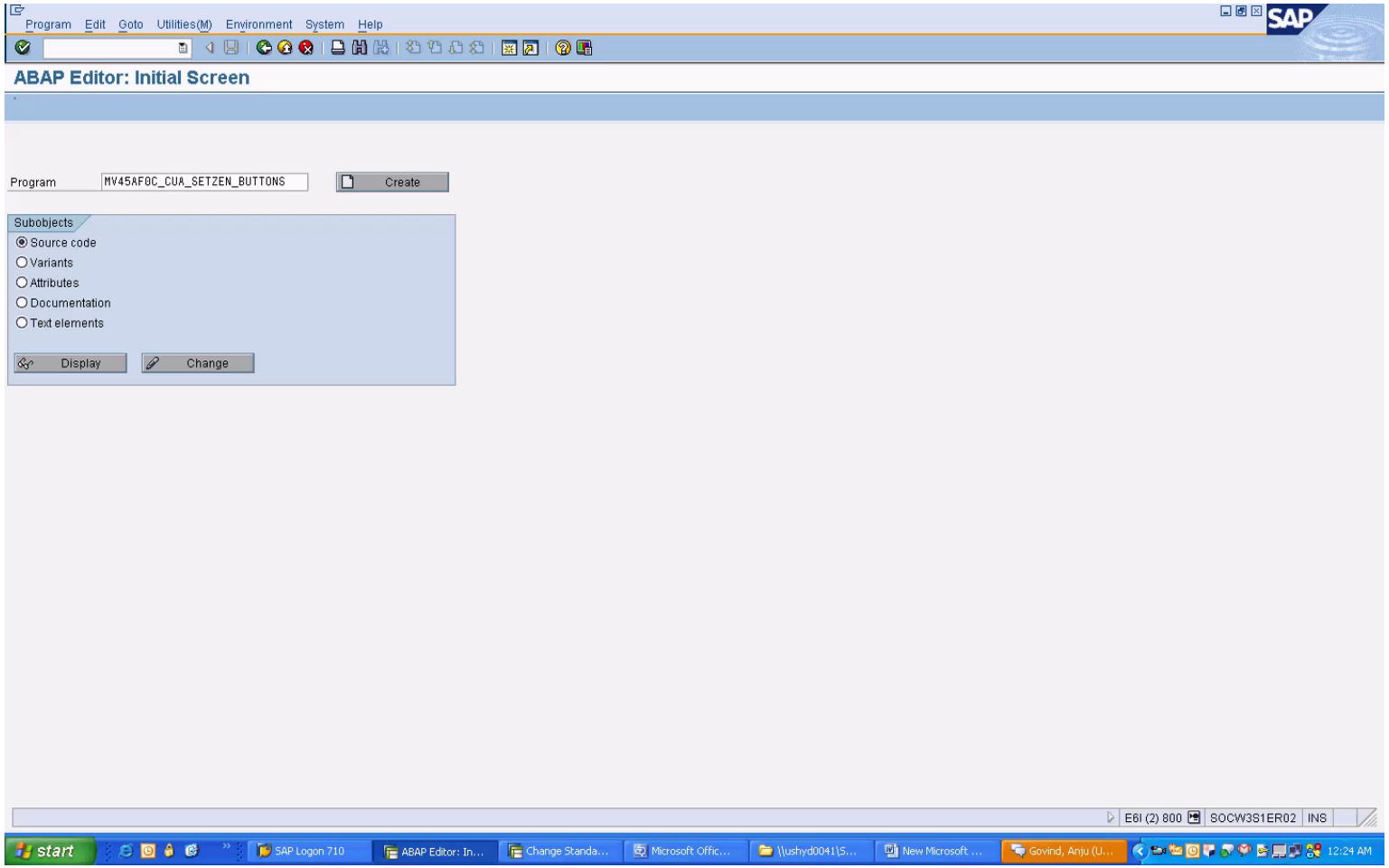
Enhancement Section—After Implementation: Demo



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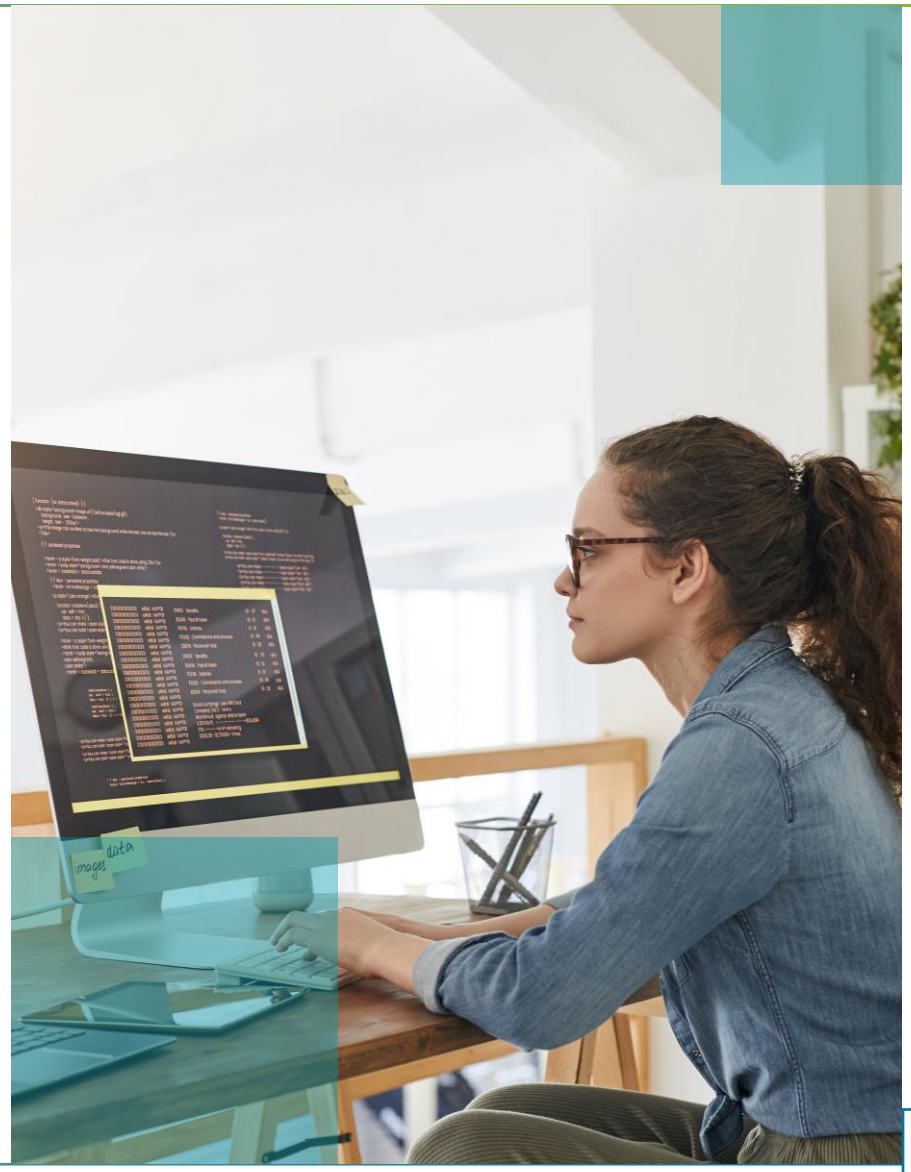
Source Code (Specified Points) Options

These are enhancement options at predefined points in a program:

- At the end of all the programs (includes, reports, function pool, module pool, etc.), After the last statement
- At the beginning and end of all FORM subroutines
- At the end of all function modules
- At the end of all visibility areas (public, protected, and private) of local class

Implicit enhancement options always exist and no enhancement spot is assigned to them.



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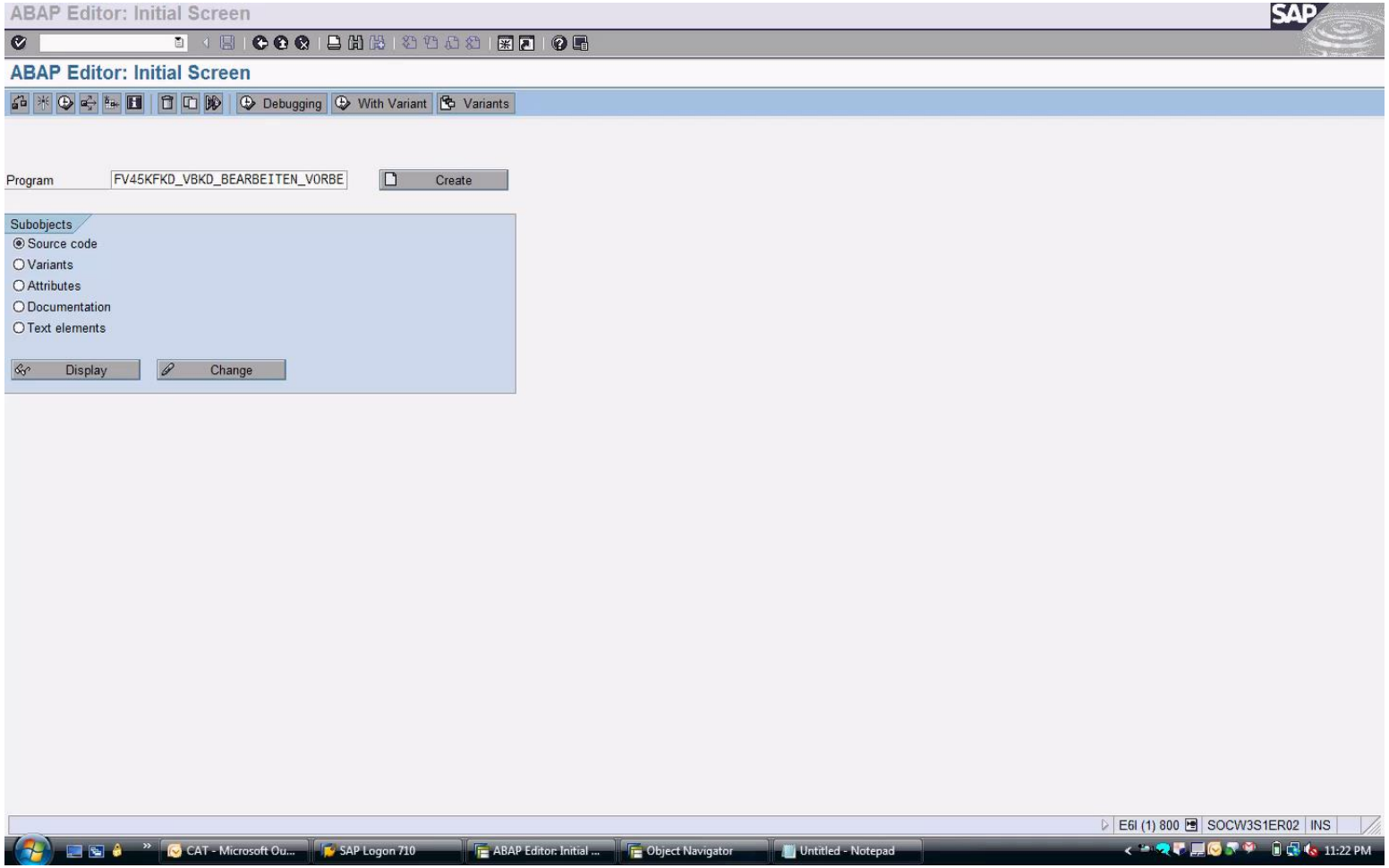
Implicit Enhancement of Subroutine: Demo



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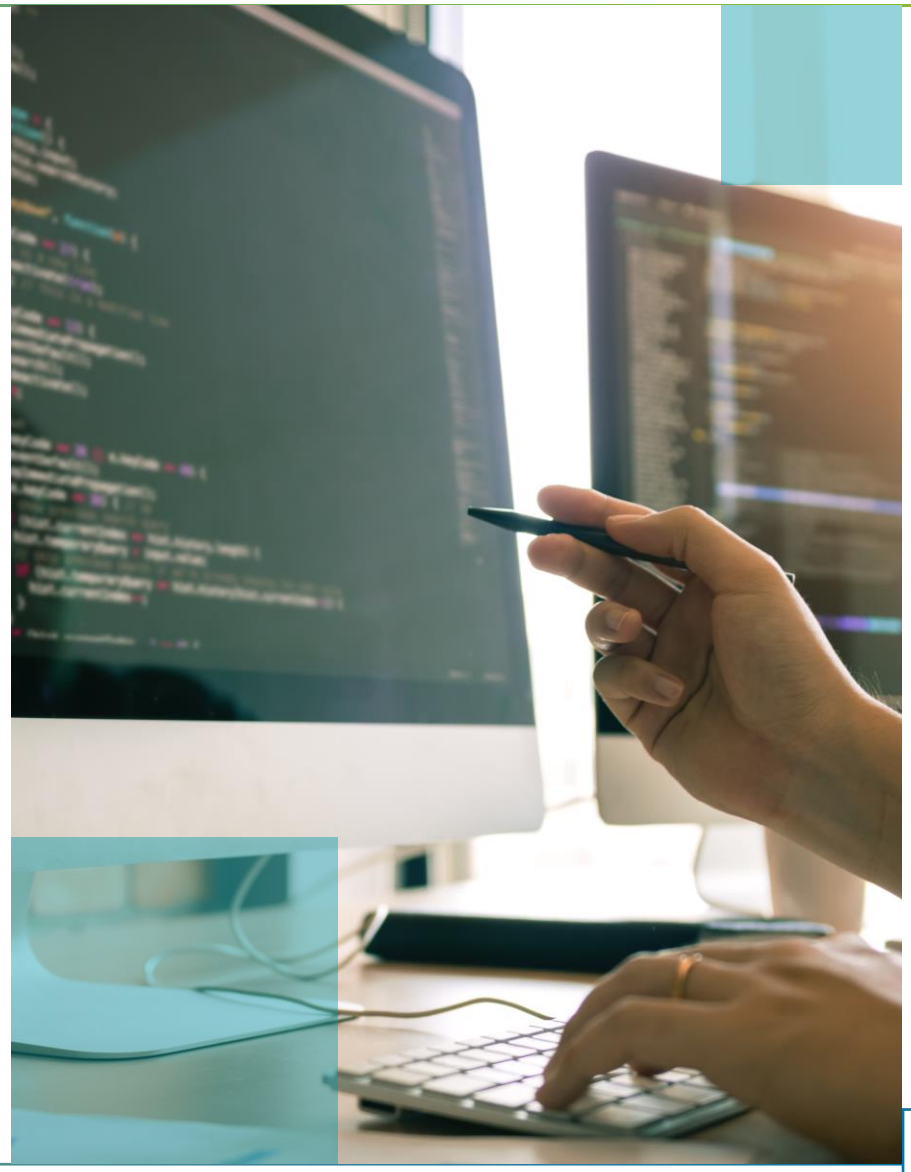
Parameter Enhancement Option—Function Group Enhancement

- The parameter enhancement option provides an option to add new parameters to a function module interface
- These parameters are always optional
- The added parameters can be utilized by making other enhancement implementations in the source code



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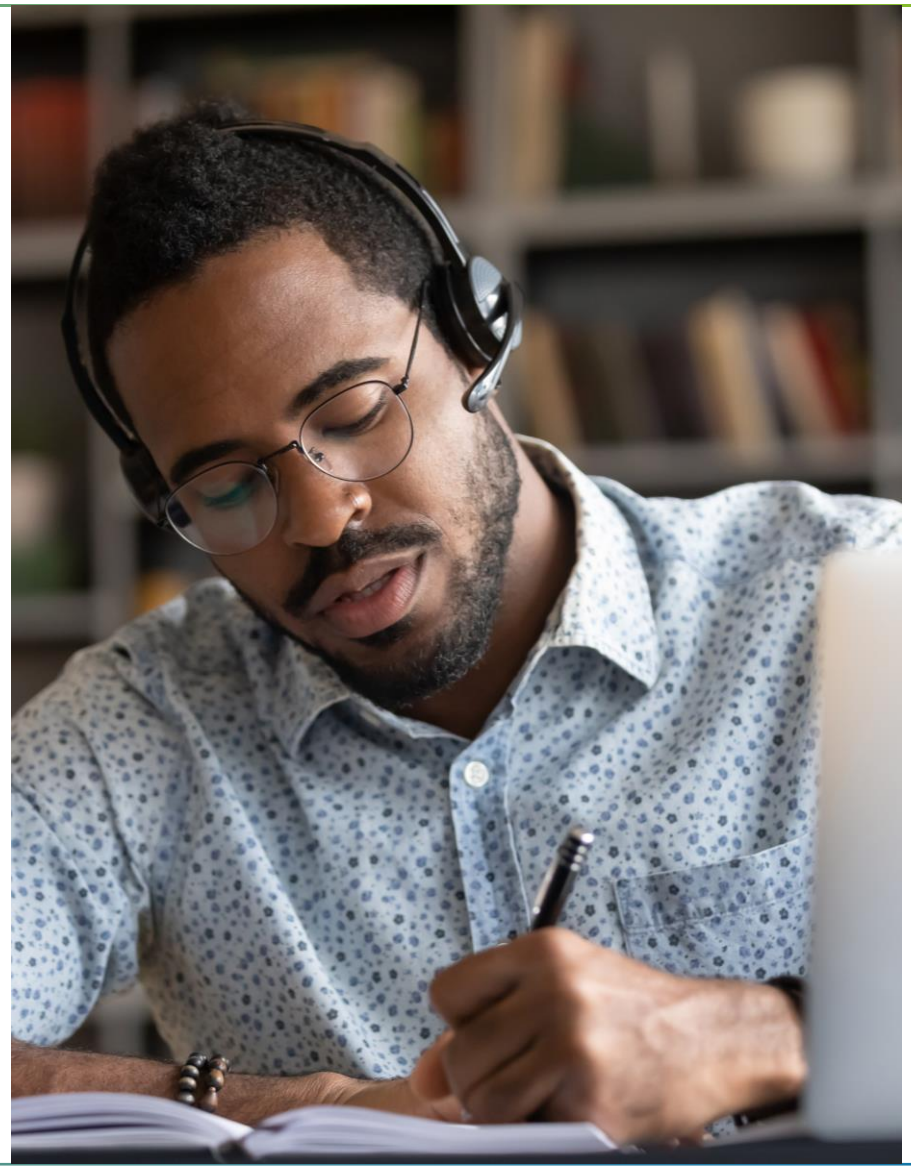
Hands-On Labs



Hands-On Activity 1

Activity Details: Enhancement spot Implementation with Material Master Transaction	
Problem Statement	Find an explicit or/and implicit enhancement during the SAVE function, while creating of purchase order through the standard transaction for material (ME21N). Implement the enhancement spot and create a record in the database table ZENHANCE##. Develop the program logic within the relevant spot, to create a record in the database table ZENHANCE##. (The value of field DESIGN = 'SPOT'.)

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Hands-On Activity 2

	Activity Details: Enhancement spot Implementation with MIGO Transaction
Problem Statement	Find an explicit or/and implicit enhancement during the SAVE function, while creating Goods Receipt through the standard transaction for Good Receipt (MIGO). Implement the enhancement spot and create a record in the database table ZENHANCE##. Develop the program logic within the relevant spot, to create a record in the database table ZENHANCE##. (The value of field DESIGN = 'SPOT'.)

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Hands-On Activity 3

Activity Details: Enhancement spot Implementation with Purchase Requisition Transaction	
Problem Statement	Find an explicit or/and enhancement during the SAVE function, while creating of purchase requisition through the standard transaction for purchase requisition (ME51). Implement the enhancement spot and create a record in the database table ZENHANCE##. Develop the program logic within the relevant spot, to create a record in the database table ZENHANCE##. (The value of field DESIGN = 'SPOT'.)

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Summary

Here are the key learning points of the module.

- The enhancement framework is a new paradigm to bring all enhancement techniques under one roof.
- An explicit enhancement option is always referenced using the spot name and option name.
- An individual enhancement inside an implementation is known as an enhancement implementation element

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